

1(a)(i)	any one from: use coloured filter / set wavelength (of light) ; calibrate colorimeter / set colorimeter (absorbance) to zero ; use distilled water (to, calibrate colorimeter / set to zero) ;	1
1(a)(ii)	1 5 stated concentrations and units ; 2 (reasonably) evenly spaced ; 3 method for dilution shown for at least 2 concentrations other than 2.0% and 0% ;	3
1(a)(iii)	correct axis labels and orientation ; x-axis label: percentage glucose concentration y-axis label: absorbance (no units or absorbance units or arbitrary units) appropriately shaped line ; line with negative gradient	2
1(b)(i)	glucose concentration (of potato) ;	1

Question	Answer	Marks															
1(b)(ii)	any six from: 1 ref. to, same / stated, potatoes (used for investigation) ; 2 ref. to, same / stated, storage conditions (of potatoes) ; e.g. (low) temperature in dark carbon dioxide concentration 3 additional stated storage condition (of potatoes) ; 4 use minimum of 5 different storage times (of potatoes) ; 5 ref. to stated storage times (of potatoes) ; 6 ref. to, same / stated, volume of potato juice (for glucose assay) ; 7 <u>at each storage time</u>, measure / note / record, absorbance (of potato juice used in glucose assay) or <u>at each storage time</u> carry out glucose assay ; 8 repeat at least twice / 3 replicates for each storage time, and find a mean ; 9 named hazard and risk and precaution ; e.g. <table border="1"> <thead> <tr> <th>hazard</th><th>risk</th><th>precaution</th></tr> </thead> <tbody> <tr> <td>Benedict's reagent</td><td>irritant toxic (to aquatic environment)</td><td>gloves / eye protection / PPE don't pour down sink</td></tr> <tr> <td>hot water (bath) A water-bath</td><td>burns / scalds</td><td>use tongs / gloves / eye protection</td></tr> <tr> <td>potato</td><td>allergy</td><td>gloves / eye protection / PPE</td></tr> <tr> <td>knife / scalpel</td><td>injury</td><td>cut away from hand</td></tr> </tbody> </table>	hazard	risk	precaution	Benedict's reagent	irritant toxic (to aquatic environment)	gloves / eye protection / PPE don't pour down sink	hot water (bath) A water-bath	burns / scalds	use tongs / gloves / eye protection	potato	allergy	gloves / eye protection / PPE	knife / scalpel	injury	cut away from hand	6
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1(c)	any two from: 1 as gamma radiation dose increases, (percentage) volatile nitrogen compounds (in raw potatoes) decrease ; 2 (percentage) volatile nitrogen compounds (in raw potatoes), are significantly different, at different doses of gamma radiation ; 3 <i>idea that</i> no change in percentage glucose (in raw potatoes), therefore decrease in acrylamide in cooked potato chips, is not due to glucose / must be due to asparagine ; 4 <i>idea that</i> decrease in volatile nitrogen compounds at increased gamma radiation doses indicates decrease of asparagine (so less acrylamide is made in the Maillard reaction) ;	2
1(d)(i)	(-)61 (%) ; correct working ;	2

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1(d)(ii)	any three from: <i>supports:</i> 1 <i>idea that</i>, gamma radiation and hot water treatment / treatment 2, decreases the (acrylamide) concentration (of potato chips) more than, gamma radiation only / treatment 1 ; 2 <i>idea that</i>, 95% confidence intervals / 95% CI / error bars (for treatment 1 and treatment 2), do not overlap (at all gamma radiation doses), therefore treatment with radiation and hot water is significantly different from treatment with radiation alone ; 3 paired data quote (to support) ; <i>does not support:</i> 4 <i>idea that</i> hot water treatment alone decreases acrylamide concentration (of potato chips) ; 5 <i>idea that</i>, 95% confidence intervals / 95% CI / error bars (for treatment 2), overlap (at all gamma radiation doses), therefore decrease in acrylamide due to hot water treatment and gamma radiation is not significantly different from hot water treatment alone) ; 6 ref. to only one variety of potato investigated ; 7 no statistical test or should carry out a, statistical test ;	3